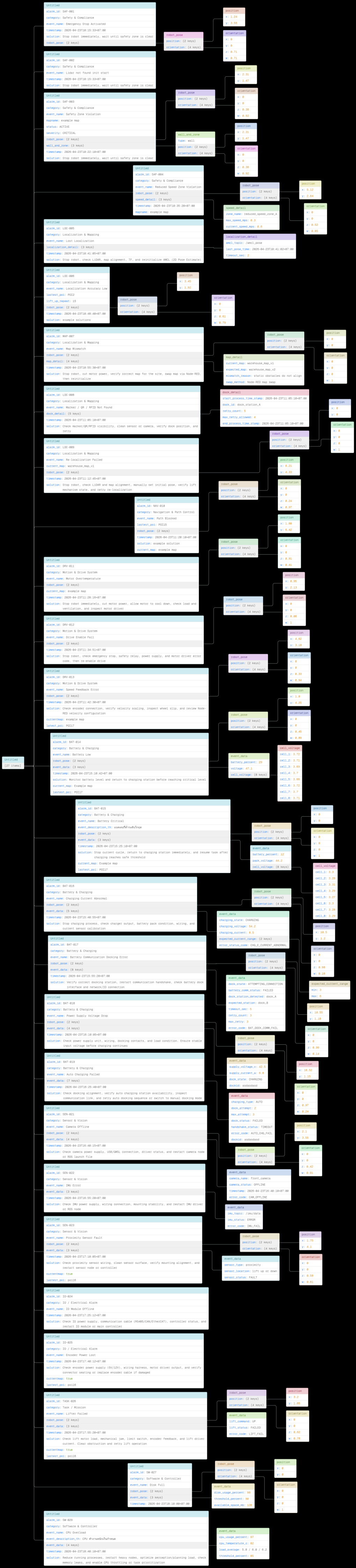
Untitled
[27 items]
Untitled
alarm_id: SAF-001
category: Safety & Compliance
event_name: Emergency Stop Activated
timestamp: 2026-04-23T10:15:33+07:00
solution: Stop robot immediately, wait until safety zone is clear
robot_pose: {2 keys}
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 1.24
y: 3.56
orientation
x: 0
y: 0
z: 0.71
w: 0.71
Untitled
alarm_id: SAF-002
category: Safety & Compliance
event_name: Lidar not found init start
timestamp: 2026-04-23T10:15:33+07:00
solution: Stop robot immediately, wait until safety zone is clear
Untitled
alarm_id: SAF-003
category: Safety & Compliance
event_name: Safety Zone Violation
mapname: example map
status: ACTIVE
severity: CRITICAL
robot_pose: {2 keys}
wall_and_zone: {3 keys}
timestamp: 2026-04-23T10:22:10+07:00
solution: Stop robot immediately, wait until safety zone is clear
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 2.31
y: 1.47
orientation
x: 0
y: 0
z: 0.38
w: 0.92
wall_and_zone
type: wall
position: {2 keys}
orientation: {4 keys}
position
x: 2.31
y: 1.47
orientation
x: 0
y: 0
z: 0.38
w: 0.92
Untitled
alarm_id: SAF-004
category: Safety & Compliance
event_name: Reduced Speed Zone Violation
robot_pose: {2 keys}
speed_detail: {3 keys}
timestamp: 2026-04-23T10:35:20+07:00
mapname: example map
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 5.12
y: 2.84
orientation
x: 0
y: 0
z: 0.52
w: 0.85
speed_detail
zone_name: reduced_speed_zone_A
max_speed_mps: 0.3
current_speed_mps: 0.6
Untitled
alarm_id: LOC-005
category: Localization & Mapping
event_name: Lost Localization
localization_detail: {3 keys}
timestamp: 2026-04-23T10:41:05+07:00
solution: Stop robot, check LiDAR, map alignment, TF, and reinitialize AMCL (2D Pose Estimate)
localization_detail
amcl_topic: /amcl_pose
last_pose_time: 2026-04-23T10:41:02+07:00
timeout_sec: 2
Untitled
alarm_id: LOC-006
category: Localization & Mapping
event_name: Localization Accuracy Low
lastest_poi: POI2
lift_up_repeat: 15
robot_pose: {2 keys}
timestamp: 2026-04-23T10:48:40+07:00
solution: example solutions
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 3.45
y: 1.92
orientation
x: 0
y: 0
z: 0.61
w: 0.79
Untitled
alarm_id: MAP-007
category: Localization & Mapping
event_name: Map Mismatch
robot_pose: {2 keys}
map_detail: {4 keys}
timestamp: 2026-04-23T10:55:30+07:00
solution: Stop robot, cut motor power, verify correct map for the site, swap map via Node-RED, then reinitialize
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 0
y: 0
orientation
x: 0
y: 0
z: 0
w: 1
map_detail
current_map: warehouse_map_v1
expected_map: warehouse_map_v2
mismatch_reason: static obstacles do not align
swap_method: Node-RED map swap
Untitled
alarm_id: LOC-008
category: Localization & Mapping
event_name: Marker / QR / RFID Not Found
dock_detail: {5 keys}
timestamp: 2026-04-23T11:05:10+07:00
solution: Check marker/QR/RFID visibility, clean sensor or camera, verify dock position, and retry
dock_detail
start_process_time_stamp: 2026-04-23T11:05:10+07:00
dock_id: dock_station_A
retry_count: 5
max_retry_allowed: 4
end_process_time_stamp: 2026-04-23T11:05:10+07:00
Untitled
alarm_id: LOC-009
category: Localization & Mapping
event_name: Re-localization Failed
current_map: warehouse_map_v1
robot_pose: {2 keys}
timestamp: 2026-04-23T11:12:45+07:00
solution: Stop robot, check LiDAR and map alignment, manually set initial pose, verify lift mechanism state, and retry re-localization
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 0
y: 0
orientation
x: 0
y: 0
z: 0
w: 1
Untitled
alarm_id: NAV-010
category: Navigation & Path Control
event_name: Path Blocked
lastest_poi: POI15
robot_pose: {2 keys}
timestamp: 2026-04-23T11:20:10+07:00
solution: example solution
current_map: example map
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 8.21
y: 4.33
orientation
x: 0
y: 0
z: 0.24
w: 0.97
Untitled
alarm_id: DRV-011
category: Motion & Drive System
event_name: Motor Overtemperature
robot_pose: {2 keys}
current_map: example map
timestamp: 2026-04-23T11:28:16+07:00
solution: Stop robot immediately, cut motor power, allow motor to cool down, check load and ventilation, and inspect motor driver
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 1.88
y: 6.42
orientation
x: 0
y: 0
z: 0.91
w: 0.41
Untitled
alarm_id: DRV-012
category: Motion & Drive System
event_name: Drive Enable Fail
robot_pose: {2 keys}
timestamp: 2026-04-23T11:34:51+07:00
solution: Stop robot, check emergency stop, safety relay, power supply, and motor driver error code, then re-enable drive
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 0.95
y: 2.13
orientation
x: 0
y: 0
z: 0.08
w: 1
Untitled
alarm_id: DRV-013
category: Motion & Drive System
event_name: Speed Feedback Error
robot_pose: {2 keys}
timestamp: 2026-04-23T11:42:30+07:00
solution: Check encoder connection, verify velocity scaling, inspect wheel slip, and review Node-RED velocity configuration
currentmap: example map
latest_poi: POI17
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 4.62
y: 3.18
orientation
x: 0
y: 0
z: 0.33
w: 0.94
Untitled
alarm_id: BAT-014
category: Battery & Charging
event_name: Battery Low
robot_pose: {2 keys}
event_data: {3 keys}
timestamp: 2026-04-23T15:10:42+07:00
solution: Monitor battery level and return to charging station before reaching critical level
current_map: Example map
lastest_poi: POI17
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 1.8
y: 4.25
orientation
x: 0
y: 0
z: 0.45
w: 0.89
event_data
battery_percent: 29
voltage: 47.1
cell_voltage: {8 keys}
cell_voltage
cell_1: 3.72
cell_2: 3.71
cell_3: 3.69
cell_4: 3.7
cell_5: 3.68
cell_6: 3.72
cell_7: 3.7
cell_8: 3.71
Untitled
alarm_id: BAT-015
category: Battery & Charging
event_name: Battery Critical
event_description_th: แบตเตอรี่ต่ำระดับวิกฤต
robot_pose: {2 keys}
event_data: {3 keys}
timestamp: 2026-04-23T15:25:10+07:00
solution: Stop current cycle, return to charging station immediately, and resume task after charging reaches safe threshold
current_map: Example map
lastest_poi: POI17
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 0
y: 0
orientation
x: 0
y: 0
z: 0
w: 1
event_data
battery_percent: 12
pack_voltage: 44.2
cell_voltage: {8 keys}
cell_voltage
cell_1: 3.3
cell_2: 3.28
cell_3: 3.31
cell_4: 3.29
cell_5: 3.27
cell_6: 3.3
cell_7: 3.28
cell_8: 3.29
Untitled
alarm_id: BAT-016
category: Battery & Charging
event_name: Charging Current Abnormal
robot_pose: {2 keys}
event_data: {5 keys}
timestamp: 2026-04-23T15:40:55+07:00
solution: Stop charging process, check charger output, battery pack condition, wiring, and current sensor calibration
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 10.5
y: 1.2
orientation
x: 0
y: 0
z: 0.98
w: 0.18
event_data
charging_state: CHARGING
charging_voltage: 54.2
charging_current: 8.5
expected_current_range: {2 keys}
error_status_code: CHG_E_CURRENT_ABNORMAL
expected_current_range
min: 3
max: 6
Untitled
alarm_id: BAT-017
category: Battery & Charging
event_name: Battery Communication Docking Error
robot_pose: {2 keys}
event_data: {8 keys}
timestamp: 2026-04-23T15:55:20+07:00
solution: Verify correct docking station, restart communication handshake, check battery dock interface and network/IO connection
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 10.55
y: 1.18
orientation
x: 0
y: 0
z: 0.99
w: 0.14
event_data
dock_state: ATTEMPTING_CONNECTION
battery_comm_status: FAILED
dock_station_detected: dock_A
expected_station: dock_B
timeout_sec: 5
retry_count: 3
max_retry: 3
error_code: BAT_DOCK_COMM_FAIL
Untitled
alarm_id: BAT-018
category: Battery & Charging
event_name: Power Supply Voltage Drop
robot_pose: {2 keys}
event_data: {4 keys}
timestamp: 2026-04-23T16:10:05+07:00
solution: Check power supply unit, wiring, docking contacts, and load condition. Ensure stable input voltage before charging continues
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 10.52
y: 1.15
orientation
x: 0
y: 0
z: 0.97
w: 0.24
event_data
supply_voltage_v: 42.5
supply_current_a: 6.8
dock_state: CHARGING
dockid: asdasdasd
Untitled
alarm_id: BAT-019
category: Battery & Charging
event_name: Auto Charging Failed
event_data: {7 keys}
timestamp: 2026-04-23T16:25:40+07:00
solution: Check docking alignment, verify auto charging station availability, inspect communication link, and retry auto docking sequence or switch to manual docking mode
event_data
charging_type: AUTO
dock_attempt: 2
max_attempt: 3
dock_status: FAILED
handshake_status: TIMEOUT
error_code: AUTO_CHG_FAIL
dockid: asdasdasd
Untitled
alarm_id: SEN-021
category: Sensor & Vision
event_name: Camera Offline
robot_pose: {2 keys}
event_data: {4 keys}
timestamp: 2026-04-23T16:40:15+07:00
solution: Check camera power supply, USB/GMSL connection, driver status, and restart camera node or ROS launch file
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 2.1
y: 3.55
orientation
x: 0
y: 0
z: 0.42
w: 0.91
event_data
camera_name: front_camera
camera_status: OFFLINE
timestamp: 2026-04-23T16:40:10+07:00
error_code: CAM_OFFLINE
Untitled
alarm_id: SEN-022
category: Sensor & Vision
event_name: IMU Error
event_data: {3 keys}
timestamp: 2026-04-23T16:55:20+07:00
solution: Check IMU power supply, wiring connection, mounting stability, and restart IMU driver or ROS node
event_data
imu_topic: /imu/data
imu_status: ERROR
error_code: IMU_FAIL
Untitled
alarm_id: SEN-023
category: Sensor & Vision
event_name: Proximity Sensor Fault
robot_pose: {2 keys}
event_data: {3 keys}
timestamp: 2026-04-23T17:10:05+07:00
solution: Check proximity sensor wiring, clean sensor surface, verify mounting alignment, and restart sensor node or controller
currentmap: true
lastest_poi: poi16
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 1.75
y: 4.1
orientation
x: 0
y: 0
z: 0.58
w: 0.81
event_data
sensor_type: proximity
sensor_location: lift up or down
sensor_status: FAULT
Untitled
alarm_id: IO-024
category: IO / Electrical Alarm
event_name: IO Module Offline
timestamp: 2026-04-23T17:25:12+07:00
solution: Check IO power supply, communication cable (RS485/CAN/EtherCAT), controller status, and restart IO module or main controller
Untitled
alarm_id: IO-025
category: IO / Electrical Alarm
event_name: Encoder Power Lost
timestamp: 2026-04-23T17:40:12+07:00
solution: Check encoder power supply (5V/12V), wiring harness, motor driver output, and verify connector seating or replace encoder cable if damaged
currentmap: true
lastest_poi: poi16
Untitled
alarm_id: TASK-026
category: Task / Mission
event_name: Lifter Failed
robot_pose: {2 keys}
event_data: {3 keys}
timestamp: 2026-04-23T17:55:20+07:00
solution: Check lift motor load, mechanical jam, limit switch, encoder feedback, and lift driver current. Clear obstruction and retry lift operation
currentmap: true
lastest_poi: poi16
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 3.2
y: 1.85
orientation
x: 0
y: 0
z: 0.62
w: 0.78
event_data
lift_command: UP
lift_status: FAILED
error_code: LIFT_FAIL
Untitled
alarm_id: SW-027
category: Software & Controller
event_name: Disk Full
robot_pose: {2 keys}
event_data: {3 keys}
timestamp: 2026-04-23T18:10:00+07:00
robot_pose
position: {2 keys}
orientation: {4 keys}
position
x: 0
y: 0
orientation
x: 0
y: 0
z: 0
w: 1
event_data
disk_usage_percent: 98
threshold_percent: 90
available_space_mb: 120
Untitled
alarm_id: SW-029
category: Software & Controller
event_name: CPU Overload
event_description_th: CPU ทำงานหนักเกินกำหนด
event_data: {4 keys}
timestamp: 2026-04-23T18:40:10+07:00
solution: Reduce running processes, restart heavy nodes, optimize perception/planning load, check memory leaks, and enable CPU throttling or task prioritization
event_data
cpu_usage_percent: 97
cpu_temperature_c: 82
load_average: 5.8 / 6.0 / 6.2
threshold_percent: 85